
P1 - E

Task 1 (Code and Chat: text-based chat, attached image) - Read through the INSTRUCTIONS.md file line by line; opened and skimmed the srs.md and felt it's a very long documents, she wanted to share to the AI; opened the Figma wireframe and skimmed the UIs design; went back to read the INSTRUCTIONS.md file and tried to understand the task; copied and pasted the srs.md contents as prompts and entered to the AI; skimmed the responses from the AI about the srs.md contents; took a screenshot of whole UI design as one image in the Figma and passed it to the AI and asked it to do a wireframe check against the image; copied and pasted what the AI replied with to the REPORT1.md without formatting the problem reports in the table; asked the AI to give her the summary regarding the problem report of SRS; then asked the AI to give her a markdown file format (without passing the REPORT1.md); copied and pasted the response for the markdown contents into the REPORT1.md file without checking;

Task 2 (Real Time: live screen sharing with voice interaction & Code and Chat: text-base chat) - read and skimmed the INSTRUCTIONS.md and tried to understand the goal of the task; followed the instructions to start the server and run the app; read the instructions in the INSTRUCTIONS.md again and decide to interact with the application; she wanted to try the Real Time this time and asked AI to help reviewing; copied and pasted the example prompt (in the INSTRUCTIONS.md) to the AI; shared her screen to the AI while interacting with the app and used the feature, and asked AI to help her find any problems; she used the application (e.g., viewed the event listing page, registered an event, etc.); asked the AI whether it can see her screen and no reply, she went back to the AI and found the screensharing is interrupted, so she resumed the screensharing; asked AI whether it can see her screen to confirm; showed the event detail page to the AI and asked it to find any issue there; continued asking whether AI can see any problem; asked AI whether it can see two Register Now buttons there; asked the AI to record the two Register Now buttons issue; went to the event listing page and used the filter and search feature and asked whether it can spot any issues; went to the INSTRUCTIONS.md file and showed the 10 heuristic table to the AI for contexts; then showed the REPORT2.md to the AI and asked the AI to report the four problems to her; copied and pasted the format of the given example in the REPORT2.md as a format reference to the AI and asked it to report the problems following the format; realized that the voice interaction feature does not work for formatted response; copied the report issues and the report format and switched to Code and Chat; pasted the contents as prompt and asked for problems report; copied and pasted the problem into the table that is not in the markdown format without checking;

Task 3 (Code and Chat: file upload and text-based chat) - read the INSTRUCTIONS.md file line by line to understand the task; browsed and opened the relative coding files .js and .css files in the file structure folder; followed the instructions to start the server, run the app, and registered events; opened and downloaded the design mockup PDF; skimmed the design in the PDF; read the INSTRUCTIONS.md again to locate the coding files and opened the MyRegistrationView.js file; used the code and chat, uploaded the design mockup PDF and asked it to make changes to the .js file to implement the feature, then copied and pasted the .js code; confirmed with AI to ask whether that updated js code is the whole script so she can replace it to the whole original version; copied and pasted the .js code into the MyRegistrationView.js; opened and copied and pasted the components.css file to the AI and asked for update for the feature implementation; copied and pasted the .css code; refreshed the app and checked the feature against the design mockup; tested the feature end-to-end against the design mockup feature;

P2 - C

Task 1 (Code and chat: text-based chat) - read and skimmed the INSTRUCTIONS.md; opened the Figma wireframe design and scrolled through; copied and pasted the example prompt and the contents in the srs.md into the AI; splitted the windows into two, left side is the wireframe and right side is the Google AI Studio; skimmed through the srs.md contents to understand the requirements and checked it against the wireframe back and forth to try to find a mismatch; went back to the AI and checked on the potential gaps/problems proposed by the AI and found that FAQ could be a problem; went back to the srs.md to locate the requirement that mention the FAQ and checked it against the wireframe; fill out the first mismatch (missing FAQ) in the REPROT1.md file himself and checked the srs.md back and forth; filled in the wrong markdown format and continued fix the wrong format; continued checked the srs.md against the wireframe design and tried to find another problem back and forth and found that it misses the text search feature; checked the srs.md and wireframe again to find the third problem, the phone number is not marked as optional; went back to the REPORT1.md file and filled in the second and the third problem in the table by himself; time is up and didn't finished the report;

Task 2 (Code and chat: text-based chat) - read the INSTRUCTIONS.md file; followed the instructions to start the server and run the application; split the windows into two with left side the app and right side the codebase repository; asked the AI to help tim review usability issues of the eventconnect app and copied and pasted the 10 heuristics table from the INSTRUCTIONS.md file to the AI for contexts; interacted with the app (e.g., browsed the event listing and event details pages); checked the contents generated by the AI; skimmed the 10 heuristics table roughly; opened the note pad to

on the other monitor to take down the problems he finds later; continued using the app, found two Register Now buttons and noted that down; tried registering an event and found a usability problem; he found that the phone number accept incorrect format and noted that down; in the registration step 3 he found that the dietary preferences selections might contradict and noted that down; tried to match the heuristic type to the problems that he found by reading through the 10 heuristic table; continued using the filter and search feature and tried to register an event again; he found that he can registered an event twice and noted that down; directly copied and pasted the formatted notes (missing some items) he wrote down into the REPORT2.md;
Task 3 (Code and Chat: text-based chat) - read the INSTRUCTIONS.md file; followed the instructions to start survey and run the app; downloaded the design mockup PDF and scrolled through to skim; copied and pasted the Background section from the INSTRUCTIONS.md file into the AI for contexts and prompted that he needs to implement a feature; split the windows into two with left is the design mockup and right is the Google AI Studio; added more contexts to the AI by describing the feature in each wireframe UIs in the design PDF in texts and checked the design mockup back and forth; copsed the pasted the .js and .css source code to the AI for more contexts to ask for update; copied and pasted the updated code into the coding source files; registered events to try to test the feature; he found the functionality is working but the style of the elements do not match the design; prompted the AI with more information to describe the changes of the visual styling and asked for update; copied and pasted the updated code to the .css file and refreshed the page to check the feature; tested the feature end-to-end;

P3 - E

Task 1 (Real Time : live screen sharing with voice interaction) - read the INSTRUCTIONS.dm file; opened the srs.md and read and scrolled through it; continued skimmed the srs.md; opened REPORT1.mf file and skimmed through it; opened the Figma wireframe design; started the voice interaction; wanted to show the Functional Requirements inside the srs.md to the AI and asked it to help her review the requirements; realized that she only turned on the voice interaction without sharing screen; she stopped the voice interaction and started live screen sharing and shared the entire screen; showed the entire srs.md and the wireframe UIs in the Figma document by scrolling through to the AI; the AI replied with one requirement mismatch (missing search bar) and she paused the live mode and went to the REPORT1.md file to write it down by herself; the live mode was stopped and she resumed it; continued asked the AI to checked the second UI wireframe design against the srs.md to find another problem; the AI replied with the 2nd problem (lacks a preferences step 2 section), and she again paused the live mode and reported that down into the

REPORT1.md file herself; resumed the live mode and the AI also found that the top navigation bar is also inconsistent and she paused the live mode and reported that down by herself; resumed the live mode and asked for the last mismatch, the AI gave her one but it seems like she don't want it and asked it to give another; the AI replied with the lack of FAQ feature across all the screens and she reported that down by herself;

Task 2 (Real Time: live screen sharing with voice) - read and skimmed the INSTRUCTIONS.md; followed the instructions to start the server and run the application; read the suggested flow in the INSTRUCTIONS.md file and interacted with the application; opened the event detail page and found that there are two register now buttons herself; refreshed the event detail page and tried to register and cancel an event; checked the About and FAQ pages quickly; used the text search feature and checked all event detail page one by one; started voice interaction and told AI that what she needed to do (she has an app prototype and she needs match whether the UIs is good and meet the 10 heuristics); want to show the 10 heuristics table and REPORT2.md to the AI; shared the entire screen and asked the AI whether it can see the 10 heuristics table, and still didn't work; she terminated the current conversation and started a new live screen sharing and asked whether it can see her screen, but the AI replied with another language other than the English; she restarted a new one and tested it; repeated the showcases of the task materials to the AI again using the screen sharing; used the application such as browsed each page, registered events; the AI replied her with the potential usability problems (two register now buttons, registration step 2 preferences multi-select options, add to calendar button is not working, and two FAQ buttons); paused the live mode and reported all problems into the REPORT2.md by writing down herself while checked the application back and forth to confirm; didn't have time to finish the last one;

Task 3 (Real Time: live screen sharing with voice, Code) - read and skimmed the INSTRUCTIONS.md ; followed the instructions to start the server, run the app, and registered events; opened design mockup PDF; read the instructions.md again and opened the source files that she needed to edit; started sharing the screen with the AI and told it that what she needed to do (implement a feature of the multiselect function) and showed the design mockup PDF by scrolling down to the AI; showed the MyRegistrationView.js to the AI and asked it how she can started for implementing the feature; the AI replied with some pieces of code, and she asked it to replied as a text so she can copy and paste; the AI replied with one line of js code, she copied and pasted the line into the js file; then the AI asked it to also applied the css style regarding that component in the js file; she asked the AI to give her the code directly and she can copy and paste it to the corresponding place in the css; after completed a couple line of code, she went back to the application and refreshed the page and found nothing update on the feature; continued the same flow with talking to the AI and added couple lines of

code in the js file; the AI kept asking her to move around the source code; refreshed the page and the checkboxes are updated in the registered event cards but wrong place; she asked the AI to how to change the code to fix the position; she updated the code generated from the AI and check the update; then she asked the AI to help her implement the Select All element by showing the design mockup to the AI; the AI replied with a chunk of code, she copied ans pasted the code in the js file and tried to fix the format; time ups and refreshed the page and found that the MyRegisteration page is blank, it seems it's borken;

P4 - C

Task 1 (Code and Chat: text-based chat) - read the INSTRUCTIONS.md upper part of the content line by line; opened the Figma design; opened the srs.md file and skimmed through it; split the windows into two with left side the srs.md and right side the wireframe design; read through the srs.md requirements line by line and checked it against the wireframe UIs one by one and tried to find a problem; tried to understand the workflow between each screen; read the use stories against the wireframe and found there's no search function in the wireframe; continued read through the srs.md and compared it to the wireframe; copied and pasted the example given in the REPORT1.md file as a markdown format; copied and pasted the functional requirements for more contexts to the AI; then he wrote down the problems behind the requirements that has problem (e.g., R1 - there should be another option for the search event in the nav bar, etc.); in total he wrote down 5-6 problems in the prompt and entered to the AI; the AI generated report in a markdown file with the problems he found; he copied and pasted it into REPORT1.md without further checking;
Task 2 (Code and Chat: text-based chat) - read and skimmed the INSTRUCTIONS.md file; followed the instructions to start the server and run and application; read the INSTRUCTIONS.md and skimmed the 10 heuristic table examples quickly; copied and pasted the the 10 heuristics to the AI as prompt and asked it to report the problems following this heuristic and the same format; split the windows into two with left the project repo and right the application; interacted with the application and used the feature; opened the event detail page and found there are two register now buttons and he reported that down under the H1 example in the prompt; tried to register events and found that there's no food options in the preferences, he reported that down under the H2 heuristic; checked the My Registration page and found that there's option he can edit the information for the registered event and reported that down under H8; he found the Add to calendar button doesn't work and reported that down under the H4; continued using the app by checking the FAQ page; tried to register another event and under the H2 he reported the problem of privacy that everyone can see the other registered attendee and the problem of payment information should be

exposed early; checked the About page and under the H1 he reported that no registration fee information upfront; he wrapped the notes and entered to the AI; copied and pasted the table generated by the AI and found that the REPORT2.md format is different from the REPORT1.md, he copied and pasted the REPORT2.md format to the AI and asked it to regenerate the table following the format; copied and pasted the table into the report without further checking; read the INSTRUCTIONS.md to make sure he did everything;

Task 3 (Code and Chat: text-based chat) - read the INSTRUCTIONS.md and followed the instructions to start the server, run the application and registered events; downloaded the design mockup PDF; read the INSTRUCTIONS.md and opened the .js and .css files; split the windows into two with the left side the project repository and right side the application; copied and pasted the js source code, the example prompt, and describe the feature he needed to implement in one sentence to the AI in the chat and entered; copied and pasted the updated js code and refreshed the page and found that the update feature is showing but not looks like the design; he checked the design mockup and described the elements of the feature back and forth;copied and pasted the js code and refreshed the page and see the feature is updated (looks good but not the same as the mockup design); continued described the changes he want to make to the AI and checked the wireframe design back and forth; copied and pasted the code and refreshed the page and checked the feature, the functionality is working, but the style is not matched; copied and pasted the css file to the AI and asked it to add some colors for those buttons and panels by describing what colors are needed matching the colors in the design wireframe; copied and pasted the the css file into the file and refreshed the page; time ups and had no time to check the update;

P5 - E

Task 1 - Read the srs.md file line by line; checked the Figma UI design; found the first requirement mismatch herself by comparing the SRS and wireframe and reported it in the REPORT1.md file; reported the mismatches by typing them herself; she mentioned that she forgot to use AI

Task 2 - Found the two “registration now” buttons by herself; used Google AI Studio Real Time feature and shared the screen in REPORT2.md, asked the AI (via voice interaction) to write down the issue in the table one by one; asked the AI to check the app’s UI usability to confirm the problem; interacted with the app by using the filtering feature and asked the AI for any usability issues; asked the AI again to help filling out the problem in the table one-by-one in a row; ask AI to slow down while writing dowm the problem (to keep the pace with her typing); asked AI to find another usability issue; when reported each usability issue: asked the AI to answer each box in a row words by words reported from the AI’s responses;

Task 3 - Used the Code and Chat feature; uploaded the Task 3 high-fidelity mockup design PDF to the AI; copied all the details for Task 3 inside the INSTRUCTIONS.md to the AI; copied all the source code in the MyRegistrationView.js file to the AI and asked for help with updating the source code; copied the wrong source code (.css code by AI) into the JavaScript file; rechecked the source code generated by the AI; copied paste the original component.css source code into the AI and asked for the updated version; refreshed the app but didn't check on the updated feature for confirmation and went to push the change directly

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P6 - C

Task 1 (Code and Chat: text-based chat) - read the INSTRUCTIONS.md line by line to understand what she needs to do; opened the srs.md file and skimmed through the content; opened the Figma wireframe; read the INSTRUCTIONS.md again about the task description; copied and pasted the example prompt into the AI and appended the contents inside the srs.md for contexts; asked the AI to give her a list of requirements that should be in the wireframe so she can have ideas on what to check on; scrolled up and down and tried to locate the latest conversation; skimmed the responses of the check list generated by the AI; split the windows into two with left is the AI and right is the wireframe; started checking the list of requirements (with checkboxes provided) from the AI against the wireframe design in the Figma back and forth to find mismatched; marked and checked those that meet the requirements and leaved those unselected that doesn't match the requirements; she found that the wireframe design is missing the FAQ feature and tried to report the problem in the table but didn't finish; she typed the issues (e.g., missing the FAQ and text search features, Sold Out event still shows register now button, and optional label is not showed in the registration flow) that she found herself and asked the AI to generate the report following the format, then she copied and pasted the REPORT2.md to the AI; copied and pasted the table as the Markdown format into the REPORT1.md and briefly skimmed through the contents to confirm;
Task 2 (Code and Chat: text-based chat) - read and skimmed the INSTRUCTIONS.md file to understand the task; followed the instructions to start the server and run the application; read the INSTRUCTIONS.md again for more information; copied and pasted the example prompt, asked it to help review the 10 heuristics, copied and pasted the the REPORT2.md file, and told that AI the she needed to find 4 problems; followed the suggested flow to interact and use the application (e.g., registered and cancelled an event, refreshed the event detail page, checked the sold out event page, registered events again); she has not found any issues upon those actions; checked the event detail page and found the two register now buttons, she went back the AI and recorded the problem in the prompt; checked the FAQ page and clicked and expanded those questions; tried using the text search and filter features to

find a problem; reported the problem that she think it's a problem (but it's not) of the text search returns irrelevant results; she went back to the text search feature again to double check and found that the text search actually returns the correct events, so she moved the problem from the prompt; continued using the app via browsing the event detail page and went to the event detail page and found the Add to Calendar button doesn't work; tried to register an event and found that the preference tab for registration for the Parking Needed as a required answer but she can bypass without choosing one (which it' incorrect) and recorded it down to the AI; entered the first two problems to the AI; copied and pasted the generated table as a Markdown format into the REPORT2.md file; time ups;

Task 3 - downloaded the design mockup PDF; split the windows into two with left if the project repo and right is the design mockup; read the INSTRUCTIONS.md file line by line to understand the task; followed the instructions to start the server, run the app, and registered events; read the INSTRUCTIONS.md file and located the two source coding files that she needed to edit; discard the old chats and started a new code and chat conversation; copied and pasted the example prompt, .js and .css source code to the AI for contexts; told the AI that she need to implement the following in the My Registration page: started describing the feature elements in details and checked the wireframe design screen by screen while elaborating the details; continued added more description as an item list (1, 2, 3, …) and checked the UIs wireframe back and forth; copied and pasted the update code into the .js file; control F to find the corresponding .css section for the MyRegistration page and pasted the updated code; refreshed the page to check the update and used and tested the app end-to-end and checked it against the design mockup and found some mismatch; prompted the AI about the change she wanted to make by checking on the design mockup; copied and pasted the updated .js file and refreshed the page and had no time left for checking; time ups;

P7 - E

Task 1 (Code and Chat feature for help writing down the problems following the markdown table) - checked and read aloud the INSTRUCTIONS.md; checked and read aloud of the srs.md document. Read aloud the requirements and checked the wireframe back and forth, and found out the first problem (missing text search feature); reported and wrote down the problem in the table; continued reading aloud section 2.3 functional requirements; found the next requirement problem himself by checking the srs.md and wireframe; asked the researcher whether he could use AI to help with writing the report1.md for the markdown table (yes); copied pasted the report table to the AI and them wrote down the problem himself regarding the problem that he found and asked AI to help him complete the problem report based on the info he entered; he went back and read the srs against the wireframe and tried to find the third problem by himself (missing the FAQs feature); he described and entered the problem for the AI

and asked AI to help him filling out into the table for quick formatting; continued read the srs against the wireframe and tried to find the last problem by himself (inconsistent nav bar); he described and entered the problem for the AI and asked AI to help him filling out into the table; waited for AI to complete the problem report in the .md table and copied paste them back to the report;

Task 2 (Code and Chat feature for help writing down the problems following the markdown table) - read aloud the INSTRUCTIONS.md file to understand the task; followed the steps to start the server; followed the steps to interact with and use the features (searched for an event, registered an event; checked the registration list; cancelled an event; tried the edge cases during the registration by refreshing the page) of the app; went through each heuristics one by one in the table and checked them against the app; found a heuristic problem (clicking the FQA button interrupts the registration process) by himself; entered the problem into the AI as prompts and asked AI to fill in the problem in the report table; went back to the EventConnect app and found the second heuristic problem (two register now buttons); enter the problem to the AI and asked it to fill in the new problem; went back to the app and interacted with the app again and tried to find the next problem; found the third problem (no closest events options in the filter feature); entered the problem into the AI for records; went back again to check the app by interacting with it and tried to found the last problem; he found out that there is an event that he thought he didn't enter any info for the registration steps before (but he did sometime before), and he thought that's a problem; he entered the "problem" into the AI for the last one in the report;

Task 3 (Code and Chat feature for uploading the design mockup as well as describing the feature's elements into the AI for more information) - read aloud the info in the INSTRUCTIONS.md file; checked the feature design mockup; followed the instructions to start the server and registered a couple of events; followed the instructions to open the files that he needs to edit; uploaded the design mockup PDF to the AI and told the app that he needs to implement a feature looks like the design; went back to the design mockup and checked every screen one by one; aftering checking the design mockup: continued the previous prompts and added more info to the AI by describing the what the elements should have for the feature by describing the design screen by screen back and forth; read and checked the design mockup carefully and recorded the new info into the AI promots; hitted the prompts and waited for the responses from the AI; copied and pasted the source generated by the AI into the project; checked whether the added code makes the feature work in the app; after checking, found out that the feature is functional working but design is a bit off (styling is not 100% matched); went back to the Google AI Studio and entered more information to the AI to tweak a bit on the feature design; adopted the new change code from the AI into the project source code and found that the design was worse than before and he decided to reverse the code to the last version; remodified and tweaked the last prompt

and hitted to the AI for new response and copied paste the new responses; found out still not working; open the inspector window and found that element that has the problem and used [some tool] to check the color hex code for that element; applied the color into the .css file himself to fix the mismatched color problem;

P8 - C

Task 1 - Split the srs.md and Figma design with split view, and read the srs.md document and checked it against the wireframe design; checked the wireframe screen by screen against the requirements (Use Case) line by line; read aloud the requirements one by one and carefully checked them against the wireframe screen by screen; copied and pasted all the text components of the wireframe design into the AI and asked the AI to check these text against the srs.md and waited for the AI responses; after getting the responses: checked the problems reported by the AI and found that some of the problems that he found previously didn't covered by the AI; added additional AI problem into to the AI and waited for regeneration; copied and pasted the content by the AI into the report1.md and checked the problems one by one to make sure it's correct;
Task 2 - read the INSTRUCTIONS.md and checked each of Nielsen's heuristics and tried to remember each one cause he had never heard of these before; thinking about giving the source code to the AI to ask for usability issues, but there's too much source code for the whole project; he interacted with the app and found that there are two "Register now" button in the event detail page; he found the source code related to the event detail page and copied & pasted the source code for that page to the AI and described the two "Register Now" buttons to the AI; asked the AI to find the issues with the app; continued using the app and tried to find the features with usability issues; found these problems (the "add to calendar" button is not working; the MyRegistration event can't be clicked with more info; etc.) but not sure about what these problems violate which heuristic; copied and pasted the Nielsen's 10 heuristics and described the problems to AI and asked it to complete the table; copied and pasted the AI's answers to the report, checked each problem one by one; tried to find another problem within 3 minutes; reported a new problem to the AI and waited for the responses; copied and pasted the AI's answers to the report, checked each problem one by one;
Task 3 - read the INSTRUCTIONS.md file and followed the instructions to start the server, opened the design mockup as a split view, and opened the required source file; brainstormed how he could use the AI for this task without uploading the PDF to the AI; he decided to describe the feature in text to the AI as prompts; checking the mockup design, the MyRegistration page in the app back and forth and continued added details to the AI regarding the feature; after described the first screen of the design mockup to the AI, he copied and pasted the source code to the AI to ask it to help implement it; meanwhile waiting for the first screen design, he decided to try to implement the design

components in the second screen of the mockup design by himself; copied and pasted the updated source code by the AI to the project and checked the updated UI by himself; he found the design didn't follow the UI mockup exactly; he located the source code component himself and fixed the issue; he continued to describe and add more deatils for the second screen to the AI as prompts but found that cumbersome (takes time); meanwhile he tried to find the corresponding element components in the source code and tried to implement part of the feature by himself; he tried to describe the design in screen 3 & 4 of the mockup to the AI, getting the color of the a specific element of the design the passed it to the AI for more details; last second to copied and pasted the updated source code and had no time to check the update on the UI

P9 - E

Task 1 -
Read the instructions.md line by line
Use the Real-Time feature
Show AI the srs.md content, scroll down the srs.md to the AI, asked AI whether it can see all the content in srs.md.
Switched to Figma, showed the AI the UI's screen by screen, ask AI to spot the issue
AI answered with 4 problems
Ask AI about the first problem again to confirm.
Ask AI whether the AI can check Report1.md and asked whether the AI can help fill in the problem in the report
Copied & pasted the items generated by the AI one by one to the table.
Tried to upload a file in the Real Time Feature, but it failed
Restarted the Real Time feature, went over the srs.md & wireframe again with the AI, and asked the AI to spot 4 problems
Ask AI to write down the problems into the formatted table and
The AI reported 4 requirements to the participant; he copied and pasted them again, one by one, into the table for each problem
The format is not correct, & switched back to the Code & Chat feature and gave the AI the Report1.md file to ask it to help correct the format
At the end of the task, he said: I wish to do the task myself if I have more time

Task 2 -
Read the instructions.md roughly
Use the Real Time feature, show the app to the AI, and ask for help spotting the issue
The AI asks him to try some features
He tried to use the Search & Filter feature and found that searching "music" didn't return anything, but it should because there is an event categorized with “music”

He continued to use other features (registration, event detail pages) and found these problems:

- 2 register buttons
- phone number is not being checked
- registration without payment?
- Add to calendar doesn't work (he thought maybe it isn't implemented yet)

Checked other features, — cancel an event immediately (cancel; about; FAQs) — delete an event
Showed the Report2.md file and format to the AI, and asked the AI to report each problem row by row
Asked AI to give all problems at once
He copied & pasted all the problem.
Downloaded the report2.md and use Code & Chat to ask AI to format the problem in the table
Have two Google AI Studio open, one for Code & Chat; one for Real Time.
Copied & pasted the generated table

Task 3 -
Read the Instructions.md file roughly & downloaded the design PDF; read the run the app; registered the events (2-3)
Continue read the instructions.md, opened the source code files
(Not sure whether the Real Time screen sharing can help that much) Use the Code & Chat feature for this task
Downloaded these files: instructions.md; Cancel_multiple.pdf; MyRegistrationView.js; Components.css
Uploaded all the files to the Code & Chat files feature, and ask AI to follow the design in pdf and to fill out the .js & .css to complete the feature.
Checked AI generated code roughly by scrolling up & down
Asked AI to comment on the added code so he can better trace it.
Copied pasted the .js content, check the updated .js file by its comment for the component
Same step for checking the .css.
Refreshed the app. Checked the new feature against the design PDF screen by screen.
Found that the "Select all" box element shows a "check" symbol instead of a "hyphen" symbol.
Screenshot the correct components to the AI, then ask for update.
Ask AI to fix the elements via describing the changes as prompts. (What should be changed to what, etc.)
Waiting for AI to generate the code.

Copied & pasted the updated code, refreshed the app via using the multi-cancel feature again.
Check the feature end-to-end to make sure it works and looks like the design mockup

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P10 - C

Task 1 - Checked the instructions.md file and asked the researcher: “Is it checking the srs.md against the Figma design?”; checked with the researcher whether he can split the windows into two; split the windows into two: left side srs.md and right side the Figma wireframe; checked the wireframe one by one to the srs.md doc by himself; found one problem (the registration form is missing the preference step); found another problem (the sold-out event still shows the “Register Now” button); found another proble (the inconsistent nav bar in the screen 3); reported the first problem (no text search field) in the table by himself; checked the wireframe and srs.md back and forth to confirm the items in the table’s row; follow the same steps to report the 2nd problem by himself via checking the srs.md and wireframe back and forth; the same steps to report the 3rd problem but only complete the first half
Task 2 - read the instructions.md file; asked the AI to shorten the explanation of each heuristic so he can recap each one; read through each heuristic generated by the AI; asked the AI which heuristic is the most common one that people often forget to implement; AI recommended the “Error Prevention”; he tried to search an non-existed event in the search text field, it works; went back to the AI to check the 2nd most common forget one “Help and Documentation”; went back to check the app; asked the AI again by providing specific context to the AI (“I’m reviewing an event registration app, give me more possible…”) so the AI the provide some suggestions to him regarding the app to check the usability; according to the suggestions from the AI and he went back to the app to check the other features; he found that the “Add to Calendar” button doesn’t work; he checked the suggestions interactions by the AI and went to the app to try back and forth and tried to found more problems; started to report the first problem (two “Register Now” buttons) in the report by himself; same way to report the second problem (“add to calendar” doesn’t work); went back to the app and tried to find other usability issues but didn’t find more since time-ups
Task 3 - read the instructions.md file and follow the steps to prep the start of the task (started the app server; opened the related files); went to the AI and provided some context to the AI about the required implemented feature by describing what the feature is about; asked the AI to help implement the “multi-cancel” feature; checked the other components’ source code in the .js file; copied and pasted the relevant source code to the AI to provide more context; checked the updated source code generated by the AI before using it; asked the AI to help implement the multi-select buttons first and checked the design mockup back and forth by zooming in and out of the elements for the feature;

tried to locate the elements inside the MyRegistrationView.js and understand the components inside the js file; copied parts of the js source code to the AI and asked it where he could start implementing the feature; skimmed the code and summary generated by the AI to tried to understand the updates that the AI made on the source code; copied and pasted the update code into the .js file and refreshed the page to check the feature against the design mockup in the PDF; then he found some mismatches between the updated feature and the design one; prompted the AI about the changes he wanted to make by describing the updates on the elements and checked the design mockup back and forth; read and checked the update code and summary generated by the AI and asked why it needs the failedcount and what does it need for to make sure he understand the update; copied and pasted the update code to the files and refreshed the page to check the feature; checked the update feature against the design mockup; time ups;

P11 - E

Task 1 - Copied and pasted the srs.md to the AI and screenshots the image of each UI wireframe design; asked the AI to check the mismatches; after AI generated the problems, she cross checked each problem mentioned by the AI by comparing them to the wireframe and srs.md; after confirmation, she reported the problem in the table by writing it herself; followed the same step to cross check the 2nd problems provided by the AI and then reported it in the table; followed the same step to cross check the 3rd problems provided by the AI and then reported it in the table; followed the same step to cross check the 4th problems provided by the AI and then reported it in the table;
Task 2 (use the speech-to-text text prompt) - followed the steps in the instructions.md to run the app, played around with the app by registering two events; asked the AI about the 10 heuristics; gave the AI the link to the EventConnect app and tried to ask the AI to check the usability; continued using the app and found some usability issues; reported the 1st problem in the table by herself (all screen nav bar doesn't have a back and reload button); reported the 2nd problem in the table by herself (two "Register Now" buttons); continued use the app and found the problems (the availability filter options show returned the sold out event) and checked the 10 heuristiv again and try to find the corresponding one; reported the last two problems following the same steps to write down by herself;
Task 3 - read the instructions.md and followed the steps to run the app and open the design PDF file; located the .js and .css files and copied and pasted the source code; copied and pasted the content in the INSTRUCTIONS.md file and the design PDF to the AI to provide more content; copied and pasted the updated source code to the AI; checked the updated UIs and found that the UIs looks different (the stylesheet source code for the app was removed); she reversed back the code and asked the AI to update

the changes that she needs without changing others; copied and pasted the updated source code to the source code files; checked the updated feature end-to-end;

P12 - C

Task 1 (didn't use AI at all) - at the beginning, he said that "based on the logic of the tasks, I'm probably choose to use the AI for the last task but not the first two tasks since he can't upload any image/video files); split the windows into two (left: the srs.md and right: the wireframe) and checked the problem manually; he found the first problem (missing the text search field); went to the Report1.md and checked the format he needs to report; followed the format and checked the srs.md and wireframe back and forth to report the first problem by himself; followed the same steps, he found the second problem (the inconsistent nav bar in screen 3); reported the 2nd problem following the same steps by checking the srs.md and wireframe to report those items; checked the srs.md and wireframe again and found the third problem (the step 1 field boxes are not marked required for all); followed the same step to report the problem in the table; same approach (check the srs.md and wireframe) and tried to find the last problem (missing FAQs); reported the problem in the table by himself;
Task 2 (didn't use AI at all) - read the INSTRUCTIONS.md Nielsen's 10 heuristics table and get familiar with the concepts because he had never heard about them; used and interacted with the app and found the first problem (the two "Register Now" buttons); checked the 10 heuristics again and tried to confirm which H# is that problem violating; reported the first usability problem himself; went back to the app and tried to register an event by typing the info in the first step, after entering all the info in this step, he decided to enter an invalid phone number and to check whether it will check the invalid phone number, turned out it went through the next step without a validation; reported the second problem in the table by himself; continued using the app to finish the registration and cancelled an event, and continued registered another event and found that found that asking the preference for each event is superfluous and he thought it's unnecessary; reported the problem in the table; continued to use the app and tried to find the last problem via using the filter and search features and found the last problem (the filtering feature doesn't allow multiple categories); reported the last problem in the table;
Task 3 (use the code and chat feature) - skimmed the INSTRUCTIONS.md file and followed the steps to run the server, downloaded the required files; prompted the AI with some context about what the feature should have and attached the two relevant source code files to the AI; checked the design mockup again and tried to get the ideas of the implemented feature; added more content into the AI for more context about the feature (for the screen 1 & 2 in the design mock); continued to check the design mockup and added more context to the AI by typing more info back and forth; entered the prompt

and wait for the response from the AI; copied and pasted the updated source code to the files and found he pasted the source code to the wrong .css file (main.css instead of the components.css) and he reversed; checked the updated feature and found the current version looks good; continued to add more context about the other elements (screen 3 & 4) to the AI by typing and describing more info in details (including the color, layout, textual content, etc); waiting for the AI generating the response & copied and pasted the updated source code into the files; while waiting he committed the interim work for backup; after updating the source code, he went back to the app to check the updated feature end-to-end; fixed the order of two buttons **(###)** by himself in the .css file using some external tool called **(###)**; checked back the design mockup to make sure the feature meet all the requirements;

P13 - E

Task 1 (use the Real-Time and Code and Chat feature) - read the INSTRUCTIONS.md line by line; used the Real-Time feature and opened the srs.md file and the wireframe UI design in the AI; asked the AI to check the srs.md and Figma wireframe design and found the problem (missing the text search field); stopped the AI screen-sharing and the live voice interaction because he found it interrupting; reported the first problem in the table by himself by checking the srs.md, report1.md format back and forth; resumed the real-time feature and asked it to check the second screen; the AI reported (a wrong problem) that the event detail page event doesn't contain the event category; participant decided to check whether that's correct or the AI gave the wrong information; he found that there's a category tag in the event detail page; he found out that the time is passing and decided to try something fast by using the code and chat feature; he took screenshots of the image of the wireframe design and copied and pasted the srs.md into the prompts and asked AI to check against the content; the AI pointed out that missing the FAQs feature and he checked the wireframe to confirm that it's correct; reported the second problem in the table by typing himself; followed the same step to screenshot the screen 2, 3, 4 of the wireframe UIs into the AI and asked it to find the problems; after the AI generated, he read through the content generated by the AI and confirm the problems are correct; found the third problem and wanted to report the problem in the table but time ups;
Task 2 (use the code and chat feature) - read the INSTRUCTIONS.md file line by line; copied and pasted Nielsen's 10 heuristics to the AI and asked it to help check the usability issues of an app; meanwhile, he went to the app and used the features and tried to find some problems; screenshot the keyword search UIs into the AI and asked it to spot any usability issue in the screenshots; copied and pasted the report2.md file to the AI and asked the AI to help him fill out the first problem; continued to use the app by registering an event and went to the My Registration page, and he found that the

registered event can not be edited; described the potential problem to the AI and asked whether there's a problem with it; meanwhile, he continued using the app and found that there are two "Register Now" buttons on the event detail page; screenshot the two "Register Now" buttons into the AI and asked the AI whether this breaks any heuristic; meanwhile, he went to check the 10 heuristics table and try to match which heuristic it violates (found it might be the H8); went back to AI to confirm with his thought and confirm it's correct; copied and pasted the problem generated by the AI into the report; went to the app and tried the filter and search feature & the register events and tried to another problem; copied and pasted the FAQs, event detail, and other pages to the AI and asked it to try to find another problem; the AI gave three potential usability problems; he was wondering which one is correct; he picked one and copied and pasted by the AI due to the time constraints;
Task 3 (use the Code and Chat feature) - read the INSTRUCTIONS.md file, opened the design mockup PDF, and downloaded the PDF; followed the instructions to register events; read the instructions again to double-check; prompted the AI that he is working on implementing a feature, uploaded the design mockup PDF, and took screenshots of the current My Registration page; located the required source files he needs to edit; went through the mockup design screen by screen to make sure he understood the feature; copied and pasted the .css and .js source file to the AI and asked the AI to update the source code based on the design; went back to the .css and .js source code and tried to understand the components; checked the .css code generated by the AI by each components and then copied and pasted the generate css by appending them to the existing file; checked the updated js file and he found that the generated js source code is given with "full" status, but he wants to only the changed source code that needed for the feature instead of copied and pasted the whole file to change the entire file so he don't want to break anything; asked the AI to give him the updated code only; the AI gave each component; he checked each component and tried to locate the each component and copied and pasted each one into the correct position; while copied and pasted the components, the syntax issues came and spenting time on fixing the syntax issue, and ended up with not finishing the feature when time ups;

P14 - C

Task 1 (didn'r use the AI) - read the INSTRUCTIONS.md line by line; opened the srs.md file and briefly went over it; checked the wireframe against the srs.md file and found the first problem (missing the text search bar); reported the first problem in the table by writing it down herself; continued checking the wireframe against the srs.md and found the 2nd problem (missing the preference step in screen 3); reported the second problem in the table be herself; same flow of checking the wireframe against the srs.md and found the 3rd problem (the inconsistent nar bar); reported the second

problem in the table be herself; checked the wireframe and srs.md doc again and tried to find the last problem; didn't find the last one cause time ups

Task 2 (Use code and chat for checking the heuristics) - followed the steps to run the app and read the INSTRUCTIONS.md file briefly; interacted with the app by using the keyword search feature and tried to find any usability issue; read the 10 heuristic table in the instructions.md file to get familiar with each one; continued using the app and tried to find the problem and found that there are two "Register Now" buttons; went back to the 10 heuristics table and tried to match the issue with one heuristic (H4); reported the problem in the table by typing it herself; continued using the app and found that the "Add to Calendar" doesn't work; copied and pasted the 10 heuristics table to the AI and described the issue button to the AI and asked it which heuristic it violates; AI returned the result (H1), she copied and pasted the item into the table; used the app and found the 3rd problem (no go back button in the event detail page); described the problem to the AI and asked the AI to categorize which heuristic this violates; copied and pasted the items generated by the AI to the report; used the registration form feature and tried to find the last problem and found that the event registration doesn't confirm submission; prompted the problem to the AI and asked the AI to classify the heuristic problem; copied and pasted the problem generated by the AI to complete the report; checked the report1.md file and found that she missed filling out the Redesign proposal column; she filled the last column by herself;

Task 3 (use Code and Chat feature) - read the INSTRUCTIONS.md file briefly and followed the instructions to start the server, download the PDF file, and register events; checked the mockup design file to understand what she needs to implement for the feature; opened the required source code file; prompted the AI with the description/specification of the feature and copied and pasted the .js and .css files for more context; asked the AI to implement the feature; copied and pasted the updated source code into the files; checked the updated feature against the design mockup and found some mismatches; prompted the AI for the some change on the existing feature (# of registration shown in the bar); copied and pasted the updated code and then checked the updated feature with the design mockup again, still some mismatches; described the change (cancel registration in the bottom) to the AI and asked to update the source code; confirm with AI whether she and copied and pasted the whole files for the change; confirmed and then copied and pasted the updated souce into the files; checked the updated feature and found that the bottom bar still has some mismatch; prompted the change to the AI for changing the feature by describing the issue info in details (color, buttons' text; layout of the buttons, etc.); copied and pasted the updated code; checked the feature end-to-end to make sure it works and looks like the mockup design

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Task 1 (didn't use AI) - Read and go over the INSTRUCTIONS.md briefly and open the wireframe; she said she wants to check the wireframe one by one by herself; read through and go over the srs.md file; confirmed with the researcher whether each UI design wireframe points to a user study; checked the wireframe against the srs.md and found that screen 3 is missing the My Registration tab; continued checking the wireframe and the srs.md and found that screen 1 is missing the text search field feature; continued checking the wireframe and srs.md and found that in the screen 3, the email and phone number is missing "required" or "optional," and also found that it is missing the preference step; after spotting all the problems, she opened the report1.md and started the write down all 4 problems in the report; tried to figure out the markdown file format while reporting the problems; confirmed the problems in the wireframe and srs.md back and forth and filled out the problems in the table; after filled out the problems and found that the markdown format of the table is messy and tried to fix it but time ups

Task 2 (didn't use AI) - read the INSTRUCTIONS.md and followed the instructions to start the server; interacted with the app, tried the filter feature, and tried to find any usability problems; browsed the event list; opened the event detail page and found that two "Register Now" buttons might violate usability; looked up the 10 heuristics table and matched the heuristic (H4) to that problem; reported the first problem in the table; checked the heuristic again and changed the H4 to H10 for the first problem; tried the search feature and tried to find another problem; tried the registration form feature and clicked the add to calendar button and found it doesn't work; looked up the 10 heuristic table and find a heuristic that match the not working button problem (H1); reported the problem in the table; continued interact with the app and tried to find another problem; she found that event that already be registered is not marked or highlighted; tried the registration form again; tried to refresh the page; found it's a little redundant to have two FAQs on the top and bottom; report the problem in the table; tried to register the same event again in the app; tried to report the last problem and time ups;

Task 3 (use the Code and Chat feature) - read the INSTRUCTIONS.md and followed the steps to start the server and opened the source code; checked the design mockup screen by screen and quickly went over the required source code; after checking everything, she said she decided to use AI to help her implement the feature; uploaded the mockup design PDF and the MyRegistrationView.js files to the AI and asked the AI to implement the Cancel Multiple feature directly; quickly went over the .css file; quickly went through what the AI gave; copied and pasted the updated .js source code to the AI; checked the updated feature and found the feature didn't change anything; checked the .js source code, .css component, and instructions.md back and forth and tried to find the issue; uploaded the .css file to the AI and asked the AI to also modify the .css based

on the design mockup; copied and pasted the updated css file; checked the udapted feature by cancel all events at once;

P16 - C

Task 1 (didn't use AI) - Read the instructions.md file line by line; opened the Figma design; read through the srs.md file; read the use case line by line and checked them against the wireframe design; read through the user story 1 & 2 and checked them against the UI design in the wireframe and found the problems (mismatch pricing, sold out event still shows the register now button); reported the problem in the table by writing it down himself; checked the srs.md and wireframe back and forth while reporting the first problem in the table; checked the srs.md and wireframe back and forth while reporting the second problem in the table; went back to the srs.md and read the user story 3 and checked it again screen 3 in the wireframe and found the problem (missing the preference indicator); reported the problem in the table by writing it down himself; checked the srs.md and wireframe back and forth while reporting the third problem in the table; checked the srs.md user story 4 and functional requirements against the wireframe and found the problem (missing the FAQ in all the screen); reported the problem in the table by writing it down himself; checked the srs.md and wireframe back and forth while reporting the last problem in the table; while reporting each problem he checked the report items at the top of the REPORT1.md file to make sure he reported the right description; after reporting all the problems, he went back to the srs.md to double-check the use case and user story for each problem;
Task 2 (chat prompting feature) - read through the INSTRUCTIONS.md file roughly and followed the steps to start the server; briefly went over the 10 heuristics; copied and pasted the report2.md file to the AI so it can refer to the format that it needs to report the problem; he used the app and found there are two Register Now buttons, checked the 10-heuristic table, and tried to find which heuristic it violates (H4, H5?); prompted the problem to the AI about the two Register Now buttons problem and so it kept track of the finding and report the problem in the table; went to the app to use the register form feature and cancel an event and didn't find out the next problem; tried searching feature and he thinks the search results doesn't match what he wants (that's could be an issue?); went to the event detail page and found that the add to calendar doesn't work and checked the 10-heuristic table, and tried to find which heuristic it violates (H8); wrote the problem to the AI so it kept track of the finding and report the problem in the table; checked the 10 heuristics table and read the H5 and found that the cancel an event doesn't confirm before it deletes; wrote the problem to the AI so it kept track of the finding and report the problem in the table; tried using the filter and the register features; checked the app back and forth and tried to find other usabiltiy problem; tried to register an event using the same email to see if it passes after checking the FAQ section; updated the web url on the top during the registration process using the same email and

it passed through without issues but the My Registration page only shows one, so no problem; tried to register an event so it closes the registration and checked whether it closed and then cancel an event and tried whether the event is opening again; checked the filter metrics and think that the Farthest First wording is not following the H2 heuristics; wrote the problem to the AI so it kept track of the finding and report the problem in the table; copied pasted the report written by the AI to the Report2.md; went over the report 2 very fast;
Task 3 - Briefly went over the INSTRUCTIONS.md this time and followed the steps to start the server and registered 2-3 events for the task; opened the design mockup file and checked the files that he needed to edit; copied and pasted the MyRegistrationView.js to the AI and then provided the context about the feature that he needs to implement by describing the feature in detail to the AI; checked the design mockup back and forth and added more details about each screen in the design to the AI; entered the prompt to the AI and waited for the AI to generate the response; copied and pasted the updated code into the files and checked the updated feature; found that the design (text and layout in the cancel confirmation box) doesn't match the mockup; tweaked the descriptions a bit in the prompt to the AI via checking the mockup design back and forth and asked for update and waiting for the generation; copied and pasted the updated code and checked the feature again; still doesn't match the mockup design; went back to the AI and re-described the change again for updating the feature and checked the mockup design back and forth; copied and pasted the code and checked the feature, and the design elements look good but the style is slightly off; went back to the AI and added more details to the AI for updated the feature styling; opened the inspection window and found the elements while waiting for the code generation; copied and pasted the code and checked the feature and times up;

P17 - E

Task 1 (Use Real Time: live screen sharing and voice) - read and skimmed the INSTRUCTIONS.md file; opened the srs.md and checked how long is the file; read the srs.md file for the use case and user stories; started sharing the entire screen with the AI using the Read Time feature; split the windows into two with left the project repo and right the Figma wireframe; stop split the two screens due to the small size; asked the AI to see his screen; started checking the requirements in the srs.md file against the wireframe UIs design by himself; then went back to the live screen sharing to asked the AI what the AI think; didn't realize that the AI can talk back to him in the live screen sharing mode and he turned on the laptop audio; opened the REPORT1.md file and skimmed through each item in the report problem; previously he spotted the problem that the registration form is missing the step 2, now he showed the AI of the REPORT1.md about the format and asked the AI helped him to report the problem he

found; copied and pasted the problem type from the AI into the table; went back to the srs.md and wireframe design to find the issues and user workflow impacts that needed to report in the table; he found there're too much information, so he decided to copied and pasted the functional requirements to the AI and asked it to help write down the problem statements needed; wrote down the first problem himself; checked the second problem reported by the AI; went back to the srs.md and wireframe back and forth to find the exact information that he needed to report the second problem; time ups without finishing reporting the second problem;

Task 2 (Use Real Time: live screen sharing with voice) - read the INSTRUCTIONS.md roughly line by line and followed the steps to start the server and run the app; went over the 10 heuristic table briefly before using the AI to help with the task; (he assumed that the AI is familiar with the 10 heuristics, but just in case) he copied and pasted the 10 heuristics table to the AI in the Real Time feature and mentioned that he needed to evaluate an app usability; he shared the screen with the AI; he tried to register an event through the flow and asked the AI to spot any usability issues while he went to the registration flow; (the AI sometimes spoken with the participant via Spanish); the AI talked to him while he registered an event; he asked the AI to help him find four usability problems while he is using the app; the AI talked to him in Spanish sometimes; the AI said the there's an error prevention usability violation issues; showed the detail page page to the AI and asked does it see any violation, and AI said no; he then went to the MyRegistration page and tried to cancel an event and realized that the cancelled the event without any confirmation; he asked the AI whether cancel an event without popup confirmation is a potential violation; the AI mentioned back that could violate H5; then he went to the About page and scrolled through the page and asked whether any of the things on that page violate usability; the AI repied with a couple potential usability issues; he started writing down the each violated elements in each row of the table in the REPORT2.md by himself; went back to the INSTRUCTIONS.md file to look up the 10 heursitics table and tried to match each problem with the corresponding heuristic that violated; wrote down the type of the heuristic that is violated of the problems that he think in the REPORT2.md; wrote down the issues of each problem in the table; time ups and he didn't finish all;

Task 3 (Use Code and Chat: text-based chat) - read through INSTRUCTIONS.md file roughly; followed the instructions to start the server and run the application and register two events; downloaded the design mockup PDF; went back and read the INSTRUCTIONS.md; opened the relevant source coding files; copied and pasted the example prompts to the AI and tweaked a bit of the prompt; copied and pasted the INSTRUCTIONS.md file into the AI for more contexts; uploaded the design mockup PDF for more contexts and entered; the AI asked him to send out the source coding files; copied and pasted the MyRegistrationView.js and components.css coding to the AI and entered; copied and pasted the updated code into the files; refreshed the page and

the feature is updated; tested by using the feature and checked it end to end; after checking that the updated feature worked successfully and still had 8 minutes left, he decided to go back to the Google AI Studio to ask for a detailed explanation about how it works; read through the elaborations about the change and update for each component on the feature; went back to check the wireframe design PDF again; found that he missed some UIs of the PDF but the AI implemented all the elements as required; roughly skimmed through the .js and .css files;

P18 - C

Task 1 (Code and Chat) - went over the INSTRUCTIONS.md file first; then roughly skimmed the srs.md file; opened the Figma wireframe; copied and pasted the srs.md and link to the Figma UI design to the Google AI Studio and prompted a brief context about the task; the AI said it cannot access the image and UIs in the Figma due to the private link, and he decided to follow the instructions of the requirements list by the AI against the UI design; found the first problem (inconsistent nav bar across the page); prompted AI to list the mismatched specifications and user story and he would manually checking the problems and passed the problems to the AI to record the problems; continued checking the srs.md against the wireframe one by one and tried to found the the next problem (the Register Now button in the sold out event detail page); prompted it to the AI for record; continued checking the srs.md against the wireframe design and found the next problem (mismatched price of an event in the event listing and detail page); prompted the problem to AI for record; continued checking the srs.md against the wireframe design and found the next problem (missing required/optional label in the form fields); prompted the problem to AI for record; read the example in the REPORT1.md file; entered the prompts and waited for the AI to generate the final table; copied and pasted each item in a row one by one to the REPORT1.md table without checking them;
Task 2 - read through the INSTRUCTIONS.md to understand the task; followed the step to start the server and run the app; copied and pasted the 10 heuristics table to the AI and decided to navigate the app himself and wrote down the problems to the AI for record; used and interacted with the app; went to the event detail page and found there are two Register Now buttons on the page and prompted that to the AI; tried to register an event and tried to leave the email field blank to see if there's any problem; tried to click the Add to Calendar button and found that didn't work and prompt it to the AI; tried using the search and filter feature; tried to cancel an event and found there's no confirmation and prompted the problem to the AI; he found three problems and can't find the last problem, so he went to the 10 heuristics table again to read through them in details; copied and pasted the REPORT2.md file into the AI and asked it to fill in the table based on the three problems he found; copied and pasted each item in a row one

by one into the REPORT2.md file without checking them; went to the MyRegistration page and checked the registered events and found that there are no details about the registered event and prompted it to the AI to update the table; copied and pasted each item in a row one by one of the last problem into the REPORT2.md file without checking them;

Task 3 - read the INSTRUCTIONS.md file and went to the repository to download the design mockup PDF; followed the instructions to start the server, run the app, and registered 2 events; skimmed the mockup design screen by screen; copied and pasted the File to Edit section to the AI; copied and pasted the MyRegisterationView.js and components.css source code to the AI; he said he want to update the design PDF to the AI and was banned by the researcher due to study design setting; so he checked the design mockup and prompted the details regarding the feature back and forth by adding more details; copied and pasted the updated code into the source files; refreshed the page and checked the feature by interacting with it; asked the researcher whether he needs to implement the feature that follow the exact style (as best as he can); prompted the AI more contexts about feature to make sure it covers it exactly look like the design mock and checked the feature & design mock back and forth; copied and pasted the updated portion of the code to the source files; checked the feature again via interacting with it and found slightly off for the style and feature format; prompted more contexts to the AI and found “An External Error” message and he refreshed the page and all the conversations was gone; he decided to end the task and said he was done;

P19 - E

Task 1 (Use Code and Chat) - read the INSTRUCTIONS.md file and copied and pasted the example opening prompt to the AI; uploaded the srs.md file to the AI; checked the Section 3 Functional Requirements in the srs.md against the wireframe screen by screen and found the first mismatch (inconsistent nav bar); reported the first problem for the first two items; took screenshots of all the UIs in Figma and the format for the REPORT1.md file to the AI; checked the problems found by the AI; removed the previously reported problem in the first row; read the first problem report by the AI item by item and went to the REPORT1.md to report the first problem in her own words in the table; checked the second generated problem by the AI against the wireframe to confirm the AI found the correct problem; reported the second problem in the table by re-wording her own language; while reporting the problem, checked the AI-generated problem against the srs.md and the wireframe to confirm the information; reported the third problem in the table by re-wording her own language; while reporting the problem, checked the AI-generated problem against the srs.md and the wireframe to confirm the information; due to the limited time left, she checked the last problem generated by the

AI quickly; reported the last problem in her own words without having the time to check against the srs.md and wireframe;
Task 2 (Use Code and Chat & Real Time) - briefly went through the INSTRUCTIONS.md and skimmed through the 10 heuristics; followed the steps to start the server; copied and pasted the example opening prompt to the AI; browsed each page of the web app; took screenshots of the event list, My Registration, About, and the FAQs pages to the AI; prompted the information regarding what she needed to find the usability issues based on the 10 Nielsen's Heuristics guidelines and copied and pasted the REPORT2.md file to the AI; for the first usability issue generated by the AI, she read through each item and checked it against the EventConnect app to make sure it's true; reported the first problem by re-wording in her own words; for the second usability issue generated by the AI, she read through each item and checked it against the EventConnect app to make sure it's correct; reported the second problem by re-wording in her own words in the table; for the third usability issue generated by the AI, she read through each item and checked it against the EventConnect app to make sure it's correct; reported the third problem by re-wording in her own words in the table; accidentally close the code and chat feature but still has the last problem she needed to find; instead open the Real Time feature, and shared the screen with the AI and asked AI to spot a usability issue in the event listing page; the AI mentioned three related problems that she already reported before and she asked for another one; the AI replied with voice interactive mode; she followed what the AI said and reported the last usability issue and asked the AI replied each item one by one and wrote it down in her own words;
Task 3 (Use Code and Chat, speech-to-text feature) - Read the INSTRUCTIONS.md file and followed the instructions to start the server and register 2 events; downloaded the design mockup file and two source code files and uploaded them to the AI; used the speech-to-text feature in Code and Chat and asked the AI that she needed to implement a feature in an app and asked it to guide her throughout the process; the AI generated the updated source code for the .js and .css files; before using the code, she asked the AI about why she needed this code and asked for guidance behind the source code; asked the AI more questions regarding the purposes of the souce code and where she need to put the code into the file; after the AI generated the code with reasons behind each added code block with comments, she read each new added block before copied and pasted in the files; after copied and pasted the blocks, she copied and pasted the whole .js file to the AI for confirmation that she is correct; accidentally close the AI conversation and the session is cleared; restarted a new conversation and uploaded all the relevant files again; copied and pasted the corresponding code again into the files; refresh the app and saw the update for the feature but the style is not correct; prompted the AI to fix the styling issue for the feature; copied and pasted the update code into the files and checked the updated

feature and found out that it's still not matching the design's style; asked the AI that the style is still not correct and asked the AI to follow the pdf design to updated the code; copied and pasted the updated code and time was up;

P20 - C

Task 1 (didn't use AI) - read through the INSTRUCTIONS.md line by line and followed the steps to open the Figma wireframe and srs.md; split the windows into two (left side is the srs.md and right side is the UI wireframe); read through the srs.md line by line and checked it against the wireframe screen by screen to find the mismatches; found three mismatches and went to the REPORT1.md file to report the first problem herself; went back to check the srs.md for the mismatches per requirements; reported the 2nd and 3rd mismatches herself by checking the srs.md and wireframe to confirm; checked the srs.md again to try to find the last problem; reported the last mismatch in the REPORT1.md file herself;
Task 2 (Code and Chat and text-file upload) - read the INSTRUCTIONS.md and briefly went through the 10 heuristics table; followed the steps to start the server and run the app; started interacting with the app by using the app like a real user and said she decided to use the AI; browsed the project app_task2 folder and downloaded the index.html file and uploaded it to the AI and prompted whether it follows Nielsen's 10 Usability Heuristics and to find any additional usability problems; meanwhile uploaded the REPORT2.md problem report table to the AI; checked the reported problems ("Summary Table") generated by the AI against the application; she continued to use the app and found the Add to Calendar button doesn't work and reported that to the AI for record; she also found that there are two Register Now buttons; tried to register an event; checked more via using the app (About/FAQ pages); followed the previous "Summary Table" generated by the AI and wrote down each problem (1-4) by herself into the table; switched the Markdown preview and editor views to confirm the edit in the table;
Task 3 (code and chat and text-file upload) - read the INSTRUCTIONS.md line by line and followed the steps to start the server, run the app, and register 3 events; downloaded and opened the design mockup PDF; she decided to upload the source code file and prompt the feature to the AI; uploaded the index.html file; prompted an overview of the new feature to the AI and provided background about which files she needed to edit; provided more details about the feature's elements by going through the design mockup screen by screen and elaborated very details (text, layout, and elements needed, etc.) for each screen to the AI; uploaded the MyRegistrationView.js and components.css to the AI for more contexts; copied and pasted the updated source code into the files; refreshed the app and checked the updated feature against the design in the mockup the file and found there're some mismatches for the feature; went

back to the AI and describe the first mismatch (positions of the checkbox for each registration card on the left side instead of the right side end) to the AI for changes; copied and paster the updated .js and .css code into the files; refreshed the app and found that problem is fixed; prompted the last fix (the position of the cancel pop up window and the exact text that each element needed “x selected,” “Clear”) to the AI for change; copied and paster the updated .js and .css code into the files; refreshed the app and found that problem is fixed;

P21 - E

Task 1 (Real Time Live Screensharing) - read the INSTRUCTIONS.md file line by line, used Google Translate to translate some words (due to not being familiar with the words as an international student); opened all related files (srs.md and Figma); opened the screen-sharing feature and shared all the details in the srs.md by scrolling down slowly; then shared the Figma and REPORT1.md to the AI for more context and described what he needed to do; shared screen 1, and the AI found that it misses the text search feature; did the same flow to go over and share the screen 2, 3, and 4 to the AI to find the problem; after that he asked the AI whether it remembers all the problems/mismatch and ask it help him to fill the first problem one by one; the AI talked too fast and he can’t follow for the typing and asked the AI to slow down; followed the same steps to report the problem one by one and he asked the AI to dictate slowly while he’s typing; then moved on to the third problem and asked the AI to repeat each item in the row one by one slowly; again asked the AI to repeat slowly so he can follow and type; then followed the same steps to report the last problem one by oen in the row; asked the AI to be slow down again and again and finished the last problem report;
Task 2 (Real Time Live Screensharing) - read the INSTRUCTIONS.md line by line and followed the steps to start the server and run the app; went over the INSTRUCTIONS.md and REPORT2.md by sharing it with the AI so it has the information and verbally gave more task information to the AI; Went to the event listing page and asked the AI to see if there are any issues on this page; then went to the event details page and asked whether the AI spotted any issues (but it missed it); he spotted the two Register Now buttons and asked the AI whether this is a usability problem and the AI said yes (H8), he said to AI “you should be carefully checking the page cause I might miss the usability issues”; then went to register an event workflow and asked the AI to spot any problem; after entered the information and tried to reload each step and to see whether there’s any problem with the usability; he tried to cancel an event and the AI spot that there’s missing a confirmation dialog for canceling an event and he asked the AI to mark that down; went to the FAQ and the About pages and asked the AI to check whether there’s feature/element that violating the usability and the AI reported the 4th problem; but before he started reporting the problems, he

want to go to check the search box feature and found that the sold out event's event detail button is still abled; he decided to report the problems in the table and asked the AI to be slowly dictate while he was typing one by one; reported the first problem from what AI said item by item in the first row; reported the second problem from what AI said item by item in the second row; reported the third problem from what AI said item by item in the third row; reported the forth problem from what AI said item by item in the fourth row, but didn't finish on time;
Task 3 (Code and Chat, File upload) - read the INSTRUCTIONS.md file line by line and followed the instructions to start the server, run the app, and register 3 events to start; downloaded the mockup design PDF and the related source code files and uploaded them to the AI, and asked it to help him implement the feature; but before he prompted it, he wanted to check the My Registration page in the app and went through the .js file to try to locate the component; while waiting, he went to the .js file and tried to add the element/component himself by understanding and locating corresponding parts (checkbox for the cancel); asked the AI which function is responsible for the register an event in the MyRegistrationView.js file and the AI said there's no such function responsible for that; asked the AI to update the code to add the checkbox and the "Select All" for him; copied and pasted the updated code and the checkboxes was added;

P22 - C

Task 1 (Code and Chat) - read the INSTRUCTIONS.md file line by line and pulled up the srs.md and Figma wireframe; copied and pasted the srs.md document to the AI and asked it to help check the problems without giving the Figma source directly; asked the AI what's the efficient way to approach it; the AI replied with a list of requirements checklist (requirement list with [yes/no]) for him to check against the wireframe; he split the windows into two (left google ai studio with the checklist in the prompts & right the wireframe); he then checked the requirements checklist against the wireframe design screen by screen and marked the [yes/no] when there were no or there were problems and also prompted the problems with some elaborations; entered the revised requirements checklist and waited for responses; copied and pasted the REPORT1.md's five items description to the AI and asked it to help fill out the problems based on the format; asked the AI to make it to report in the table format (requirement violated, problem type, UI screen and element, etc.); copied and pasted the whole AI-generated table as Markdown file into the REPORT1.md;
Task 2 (Code and Chat) - read the INSTRUCTIONS.md line by line and followed the steps to start the server and run the app; copied and pasted the task2's description (task 2 background, 10 heuristic tables, etc.) in the INSTRUCTIONS.md file to the AI for contexts; the AI replied and a list of app's interactions and potential violated heuristics;

he followed the AI's suggestions to interact with the app and tried to spot the usability issues; registered an event and cancelled an event (found that there's no confirmation pop up); registered an event and found whether there's a back and cancel buttons; found an event that was sold out (there's a Registration Close button);
Task 3 (Code and Chat) - read the INSTRUCTIONS.md file roughly and followed the instructions to start the server, run the app, opened the related source code files; split the windows into two (left is the codespace & right is the Google AI Studio; copied and pasted the task3's descriptions (background, etc.) from INSTRUCTIONS.md to the AI and prompted his task goal with elaboration for more contexts; downloaded and uploaded the source code files (MyRegistrationView.js and components.css) to the AI; continued added more details to the AI by describing the new feature's elements layouts/formats screen by screen in the mockup design wireframe; copied and pasted the updated the source code files; refreshed the app and found that there's no events registered;

P23 - E

Task 1 (Use Code and Chat & Real Time Screensharing) - Roughly skimmed the INSTRUCTIONS.md file and read through the srs.md and REPORT1.md files to get the overview of the task; shared screen of the srs.md file and Figma wireframe to the AI for contexts and asked it to help check the mismatches; copied and pasted the srs.md and reported 5 items in the REPORT1.md file into the chat below for more contexts; went over the Figma wireframe screen by screen to the AI; the AI only found one problem without further actions; stop the Real Time feature; use the code and chat feature and copied and pasted all the srs.md, INSTRUCTIONS.md, and screenshots each UIs in the Figma to the AI in the prompt in the code and chat for contexts; the AI responded with 4 issues into the list of items format; copied and pasted the format of the REPORT1.md table into the AI and asked it to report the problems into the table format; copied and pasted each item one by one into the rows in the table; went to the AI and prompted it to ask it to double check whether the mismatches it found was correct;
Task 2 (Code and Chat) - skimmed the INSTRUCTIONS.md file roughly and followed the instructions to start the server and run the app; used and interacted with the app and found that there are two Register Now buttons in the event details page (thinking of that would be one of the usability issue); tried to register an event and refresh the My Registration page; opened the Code and Chat feature and copied and pasted the 10 heuristics table to it for analyze them for contexts; screenshots the event listing page, event details page, and registration steps pages to the AI and asked it to find any usability issues; the AI gave 5 usability problems based on the screenshots; asked the AI what could it want him to test out; the AI asked it to test the Registration flow; he followed the suggestions from the AI to test out the Registration flow and tried to find

any usability issues; added more screenshots (About, FAQ pages) to the AI for more contexts and find more usability problems; copied and pasted each item one by one into the rows in the table; went to the AI and prompted it to ask it to double check whether the problems it found was correct and based on the images he provided;
Task 3 (Code and Chat) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server, run the app, and register events; downloaded the mockup design PDF and skimmed through the design; opened and downloaded the source code files (MyRegistrationView.js and components.css); uploaded the source code files, mockup design PDF, and example prompt (in the INSTRUCTIONS.md) to the AI and asked it to implement the feature; copied and pasted the updated code into the files; checked the INSTRUCTIONS.md file and asked AI how to check whether the feature is implemented and added correctly; followed the checklist steps generated by the AI to check the feature: refreshed the page; checked the events; de-checked the events; cancel multiple events at the same time, etc; check the feature end-to-end; screenshots the “No registration page” into the AI and asked it whether the last screen design was meet;

P24 - C

Task 1 - read the INSTRUCTIONS.md file line by line and opened the srs.md and Figma; read and skimmed the srs.md file; copied and paste the example prompt provided and the srs.md to the AI, and added more contexts by describing each wireframe pages name (Event Listing, Event Details, Registration pages) and entered; while waiting for AI response, he went to read the srs.md and tried to compare it against the UI designs himself; went back to the AI and it gave a list of requirements check; described what he saw for the wireframe designs to the AI (e.g., the first wireframe shows…; the second wireframe is you can…; the third wireframe is…; and the last wireframe is…); the AI generated with a list of requirements checkboxes, he compared the wireframe against the checkboxes (found missing a keyword search box); in the srs.md, used the control-F to find the related part that mentioning the “keyword”; reported the first mismatch himself in the table; found the second problem Register Now button is not gray/disabled for a sold out event; reported the second mismatch himself in the table; time up
Task 2 - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server and run the app; interacted with the app by using the keyword search, category search, and sorting features and tried to find usability problem; just the MyRegistrations, About, and FAQ tabs quickly; checked the event details page and found two Register Now buttons; tried to register an event and double checked the registered event’s details with the event listing page event’s details; tried to register an event twice using the same name and email and didn’t find any more problems; went to the 10 heuristics

table and read through them and tried to find problems that he just interacted with the app based on these heuristics; went to the detail page and clicked the Add to Calendar button and found it didn't work; found that the registered event doesn't have the confirmation-code (which showed right after the event registration) in the MyRegistrations page (categorized it as H5 violation); reported that missing confirmation-code problem himself in the table; tried to find the second usability issue by read through the 10 heuristics again; he found that cancel an event without any confirmation popup immediately and categorized that as violating the H9; reported that cancel without confirmation problem himself in the table; went to the app and used the keyword search feature by searching date/location and tried to find another usability problem; tried to register an event but times up;
Task 3 - read the INSTRUCTIONS.md file line by line and followed the instructions to start the server, run the app and registered 2 events; opened the relevant source code files; copied and pasted the INSTRUCTIONS.md, MyRegistrationView.js, and components.css for contexts to the AI; copied and pasted the updated code generated by the AI; refreshed the page and found the feature is updated and checked the feature against the give mock-up design, found some inconsistencies with the design; went to the AI to provide more detailed contexts by describing the changes needed to make for the feature and checked the mock up design back and forth while describing the changes; refreshed the page and found some styling mismatches; described the changes to the AI for more contexts regarding the styling; copied and pasted the code and refreshed the page found that other pages' styling were gone (because he replaced the whole components.css with only the MyRegistration page portion of the css code generated by the AI); reversed back the css code and updated the css code only for the MyRegistration page; refreshed the page and tried the feature end-to-end to make sure it works like the design;

P25 - E

Task 1 (Code and Chat & Real Time) - skimmed through the INSTRUCTIONS.md and open the srs.md and Figma wireframe design; prompted to ask the AI to help him review the wireframe against the srs.md file following the REPORT1.md format; copied and pasted the contents in the srs.md, Figma external link, and the table format items for more contexts to the AI; end the code and chat conversation and used the real time feature; turned on the Talk feature and prompt the task requirement (similarly prompt as previous including the srs.md and REPORT1.md); shared the screen to the AI for the Figma; asked the AI to find any mismatches; the AI replied with the first problem (inconsistent nav bar); asked the AI to write the problems down following the format and copied and pasted the REPORT1.md format as prompt; continue to talk to the AI back and forth to find the problems via showing the Figma wireframe; the AI returned the four

problems following the items in the REPORT1.md and not in a formatted form; he talked to the AI and asked it to return them in a format but didn't work (the real time feature replied with transcription but not like a chat feature); he copied and pasted each item one by one of the row in the table from the AI;
Task 2 (Real Time) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server and run the app; used the Real Time's Talk feature, copied and pasted the example prompt to the AI, asked the AI to help him check the application's usability issues; copied and pasted the REPORT2.md to the AI and asked the AI to report the problems later to follow the format; confirmed whether the AI can see his screen back and forth cause the Google AI Studio had some technical issues (the screen-sharing is not working properly); after a while (~2 minutes) the issues were gone; he started sharing the screen and asked the AI to help him check the usability problems of the app; he used and interacted with the AI by using the features (register events, browse the events, search and filter feature, browse the About and FAQ tabs, etc.) and showed to the AI of his interactions meanwhile; asked the AI: "what do you want me to look at right now" after interacting with the app himself; not sure the event details page contains enough details or not and asked the AI regarding this; continued talked and shared the screen with the AI to check back and forth and asked AI to keep the problems in records; after the AI helped him find the 4 problems, copied and pasted the REPORT2.md to the AI and asked the AI to report the problems to follow the format; copied and pasted the problems generated by the AI one by one into the table;
Task 3 (Real Time) - skimmed the INSRUCTIONS.md file and followed the instructions to start the server and run the app; shared the screen and talked to the AI about the task 3 goals; showed the AI for the mockup design PDF regarding the feature; browsed the source code files and copied and pasted the app.js code to the AI; asked the AI what does he need to do in order to implement the feature; the AI replied with the overview of the feature implementation; he asked the AI to write down the code to him directly; the AI replied with updated code but didn't give any code and he was confused and confirmed back and forth with the AI; he asked the AI that he didn't see the code and asked it to paste the updated code in their conversation; AI generated the code, he copied and pasted the code (one-line code with format); went to the app to register an event; asked the AI to correct the code because there's some syntax/logical issues; copied and pasted the updated .js code; refreshed the page and looks like the feature is added; copied and pasted the components.css source code to the AI and asked it to implement feature following the style; registered two events and tested feature end-to-end;

P26 - C

Task 1 (Code and Chat) - skimmed the INSTRUCTIONS.md file and opened the Figma wireframe design; read and went over the srs.md file; checked the srs.md's requirements list against the wireframe design back and forth and tried to find a mismatch; found the first problem (missing the keyword search feature) and marked it down in the REPORT1.md file; continued to check the srs.md against the wireframe and found that second problem & marked it down in the REPORT1.md file; continued to check the srs.md against the wireframe and found that second problem (sold out event is not present) & marked it down in the REPORT1.md file; continued to check the srs.md against the wireframe and found that third problem (email is not optional) & marked it down in the REPORT1.md file; continued to check the srs.md against the wireframe and found that last problem (no FAQ problem) & marked it down in the REPORT1.md file; copied and pasted the marked down notes for the 4 problems, the REPORT1.md table format to the AI and asked it to help it report all the problems following the given format; copied and pasted and edited based on the content generated by the AI for each item in the row for each founded problems into the table; after copied the first two problems, he asked the AI to fill out the table using the markdown format so he can copy and paste directly; copied and pasted the entire table and then checked and edited parts of the table's content based on his own opinions while checked the srs.md and wireframe design back and forth;
Task 2 (Code and Chat) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server and run the app; interacted with the app and used the features (e.g. search and filter,); copied and pasted the example prompt in the INSTRUCTIONS.md file to the AI and asked the AI to answer only yes or no; copied and pasted the app's text layout to the AI for contexts; asked the AI to help walk him through on finding the heuristic; problems; meanwhile, he continued using the app and found there are two Register Now buttons in the event details page and prompted that to AI; copied and pasted the REPORT2.md file to the AI and asked it don't give the proposed redesign to him; followed the AI generated suggested heuristic walk through and checked them in the app back and forth; when he found a mismatch from the AI generated suggestions list in the app, he wrote that down in the prompt for records (e.g., no dark mode; no reset feature, etc.); continued using the app by registering an event; entered the prompt; the AI generated table with the Proposed Redesign column; copied and pasted the text contents of the application to the AI and asked it to propose the redesign and cross referencing the text contents of the application; the AI generated the proposed redesign based on other metrics (e.g., cognitive workload) and he asked the AI to generate the proposed redesign based on the heuristics only; copied and pasted the table into the REPORT2.md file and no further checking;
Task 3 (Code and Chat) - copied and pasted the entire INSTRUCTIONS.md file and followed the instructions to start the server, run the app, and registered 2 events; started a new conversation and copied and pasted the example prompt to the AI; copied and

pasted the current text contents and layout of the My Registrations page to the AI for more contexts; prompted the AI to act (a role) as a senior software engineer and come up with a plan for implementing the feature, and gave more details regarding the feature by describing the feature's elements while checked the design mockup back and forth, copied and pasted the .js and .css source code to the AI for more contexts; asked the AI to double check it; asked the AI whether the generated code is the entire file for including the original source code; copied and pasted the updated code into the files; refreshed the page and the feature is updated, finished without checking whether the feature is working or not;

P27 - E

Task 1 (Real Time & Code and Chat) - skimmed the INSTRUCTIONS.mf file and opened the srs.md and Figma wireframe; copied and pasted the srs.md file to the AI and asked it to summarize the contents for him; prompted the AI that what he needed to do and asked it to help him review the wireframe against the srs.md; started sharing the screen and showed the wireframe UIs screen by screen; asked the AI to check the wireframe against the srs.md and copied and pasted the srs.md again; he switched back to the Code and Chat to do the task; copied and pasted the Figma link to the AI (but it can't access to the external link); screenshot each wireframe and copied and pasted the srs.md as prompt and asked the AI to check the wireframe against the srs.md and find the problems; the AI generate the founded problems in a list of mismatches but not in the REPORT1.md format; copied and pasted the REPORT1.md file to ask the AI to report the problems following the format; copied and pasted the table generated by the AI into the REPORT1.md; checked each problems against the previous (a list of mismatches) to double check it's correct;
Task 2 (Real Time) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server and run the application; opened the Real Time feature, and copied and pasted the example prompt and 10 heuristic table to the AI and asked it to check the app against the 10 usability heuristics, specifically prompt the AI that "Do not give the results until he said Give Me the Results"; shared the screen to the AI and started interacting with the application (browsing the events, registering an event; checking the My Registration, About, and FAQ tabs, clicking the Add to Calendar button, searching and filtering features, etc.); checked the logical issues while interacting with the app (e.g., whether registering an event successfully marked one spot taken, searching and filtering feature working, etc.); talked to the AI "Give Me the Results" for reporting the usability problems that it found of his interaction with the application; copied and pasted the REPORT2.md file to the AI and asked it to report the usability problems following the format; copied and pasted the table generated by the AI

into the REPORT2.md file; he didn't check because he said he don't know the heuristics and he assumed the AI is correct;
Task 3 (Code and Chat) - skimmed the INSTRUCTIONS.md and followed the instructions to start the server, run the application, and register 3 events; skimmed through the mockup design PDF for the feature; downloaded the PDF and uploaded it and screenshots the current application's My Registrations page to the AI in the prompt, and asked it to compare them; opened the two source files (MyRegistrationView.js and components.css); copied and pasted the two source code to the AI; he said what the AI generated to him, he just gonna copied and pasted back into the files; copied and pasted the generated code into the file; restarted the server to refreshed the change for the feature; used the updated feature to check whether it works and checked the design mockup back and forth;

P28 - C

Task 1 (Code and Chat: file upload & chat prompt) - read and skimmed the INSTRUCTIONS.md file; opened the srs.md file and skimmed the requirements; read through the use case and user stories and checked it against the wireframe design in the Figma back and forth; found the first problem (missing the keyword search feature); started to report the first problem in the first row in the table of REPORT1.md; uploaded the srs.md file to the AI and asked it to give him the functional requirements and a checkboxes list of the requirements; meanwhile he went through the user stories real quick; checked the AI generated functional requirements v.s. related user cases & user stories and wrote down the possible Requirement Violated column; finished reporting the first problem himself; continued checked the wireframe against the srs.md requirements and found additional two problems (the sold out event still shows the Register Now button & the mismatched prices between the event listing page and event details page); reported the 2nd problems item by item in the table himself; continued checked the srs.md against the wireframe design and found that the registration form screen is missing the step 2; reported the 3rd problems item by item in the table himself; continued checked the srs.md against the wireframe design and found that all screens are missing the FAQ tab; reported the 4th problems item by item in the table himself;
Task 2 (Code and Chat: files upload and chat prompts) - skimmed and read the INSTRUCTIONS.md file and followed the instructions to start the server and run the application; downloaded and uploaded both the INSTRUCTIONS.md and REPORT2.md to the AI and asked it to give him one sentence to summarize the 10 heuristics and give him a template of what the report might look like for each heuristics; meanwhile skimmed the REPORT2.md about what he needs to report; splitted the windows into two (left the AI and right the application) and tried to use and interact with the app to find the usability problems; he found that there are two Register Now buttons

in the event detail page; he wrote that down to the AI for record; continued using the application to find other usability problems (e.g., checked the event prices, registered an event, cancel an event, signed up the same event twice, used search and filter features, checked the About page, signed up multiple events, clicked the Add to Calendar button in the event details page); reported the problems to the AI in the prompt (e.g., no undo a cancel of registration, text search feature shows the events that it shouldn't, the Add to Calendar didn't work); checked the AI generated table regarding the four problems that he gave to see if it makes sense; wrote down the problems himself item by item of each row into the table by using his own descriptions after reading through AI-generated descriptions; while reporting the 1st problem, he double-confirmed whether the AI reported problem descriptions are matching the problem he found back and forth and tried to understand whether the heuristic is making sense for the 1st problem; while reporting the 2nd problem, he double-confirmed whether the AI reported problem descriptions are matching the problem he found back and forth and tried to understand whether the heuristic is making sense for the 2nd problem; while reporting the 3rd problem, he double-confirmed whether the AI reported problem descriptions are matching the problem he found back and forth and tried to understand whether the heuristic is making sense for the 3rd problem; while reporting the 4th problem, he double-confirmed whether the AI reported problem descriptions are matching the problem he found back and forth and tried to understand whether the heuristic is making sense for the 4th problem;

Task 3 (Code and Chat: files upload and chat prompts) - skimmed and read the INSTRUCTIONS.md file and followed the instructions to start the server, run the application, and register 2 events; opened the design mockup PDF and skimmed the design screen by screen and continued read the instructions in the INSTRUCTIONS.md file; downloaded the .js and .css files and prompted the AI that he needs to "adding a function that allows me to…(descriptions of the feature) and updated the source files that reflect the changes specified below:"; splitted the windows into two (left is the AI and right is the PDF); elaborated and prompted the descriptions of each screen in the PDF design for the feature in very specified details; checked the design mockup back and forth and prompted more details to the AI for more contexts; double-checked he mentioned all the changes before prompting to the AI and asked the AI to tell him what it will do visually before he implement the feature; he forgot to upload the files and immediately uploaded the two source files; copied and pasted the AI-generated code into the files; refreshed the page to see the updated feature; splitted the windows into two (left the design mockup and right the application's update feature page) and checked the updated feature end-to-end against the design mockup and found some styling mismatches; prompted the AI for more changes by describing what he wants to make; copied and pasted the updated code into the source code files and refreshed the app to check the updates he wants to make real quick; times up;

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Task 1 (Real Time feature: screensharing and text prompt) - read the INSTRUCTIONS.md file line by line and opened the wireframe and srs.md; read the srs.md line by line and skimmed the wireframe; shared the screen to the AI (using the Real Time feature) and went over the srs.md by scrolling down and also shared the wireframe UIs design screen by screen to the AI while talking to the AI to ask it to help spot the problems; the AI reported one mismatch (missing the text search field); then she showed the REPORT1.md file to the AI and asked it to report 4 problems to her; AI replied with 4 mismatches (missing text search field, sold out event shows Register Now button, missing the step 2 for the registration, etc.) based on the previous information from the screensharing; she double-checked whether the AI reported mismatches are correct/existed or not by checking the wireframe and srs.md; then she decided to fill out the problems by herself in the table as well as talked to the AI to ask for which use case/user story is violated; while found the 1st problem given by the AI and then reported the 1st problem in the table, she checked the srs.md and wireframe back and forth for each item in the problem row; while found the 2nd problem given by the AI and then reported the 2nd problem in the table, she checked the srs.md and wireframe back and forth for each item in the problem row; while found the 3rd problem given by the AI and then reported the 3rd problem in the table, she checked the srs.md and wireframe back and forth for each item in the problem row; while found the 4th problem given by the AI and then reported the 4th problem in the table, she checked the srs.md and wireframe back and forth for each item in the problem row;
Task 2 (Real Time: screensharing) - skimmed the INSTRUCTIONS.md and followed the instructions to start the server and run the app; followed the INSTRUCTIONS.md suggestions to tried using the app (e.g., opened an event, registered events, browsed event listing page, checked the About page); start sharing the screens to the AI and showed the INSTRUCTIONS.md and the 10 heuristics table for contexts; then went to the application to show the AI all the features and use the features of the application by going through these workflow (e.g., browsed the event listing page, went to the event detail page (said there're two Register now buttons to the AI), registered an events, etc.); then she asked the AI to report the four problems for her by showing the REPORT2.md format (what items she needs to report for each usability problem); the AI gave her 4 problems and the corresponding heuristics that were violated; while found the 1st problem given by the AI and then reported the 1st problem in the table, she checked the application regarding the usability problem given by the AI to double-checked is correct, reported the 1st problem in the first row; the Real Time turned down due to the technical issues and she needed to go over the application's workflow again to the AI and asked it to find the 4 usability problems again that she

previously did; the AI reported the second problem (cancel an event without a confirmation dialog/popup), she checked the AI generated suggestions and filled out 2nd problem in the table; read through the AI suggested 3rd problem in details (from the transcription) and went back to the application to double-checked whether that usability issue makes sense; asked the AI to give her the 4th problem and the AI responded with one but she said that the mentioned problem might not be counted as a usability problem because she think it's very common; she asked the AI to give her an additional usability problem; the AI gave her another usability problem and she went back to the app to check whether the usability issue makes sense and she thinks it's an usability issue; reported the 4th problem in the table;

Task 3 (Code and Chat: file uploads and text prompts & Real Time: screensharing) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server, run the app, registered 2 events, opened and downloaded source code files; opened the Code and Chat and uploaded the design mockup PDF and thinking of sharing the source code as prompts; but after thinking a minute, she changed the mind that maybe the Real Time might be easier for the task 3; she shared the screen to the AI, then she showed the original My Registration page to the AI, and then went over the mockup design PDF as well as described the feature orally in details by going over each screen; then she copied and pasted the .js and .css files to the AI and asked it suggest the code change for her; the AI gave her the line of code that she needed and she copied and pasted the code into the source file; she asked the AI whether she did it correctly, the AI consistently said she didn't do it correctly; followed the same steps to add the additional suggested code change into the .js file and double-checked with the AI everytime; basically she asked the AI to guide her to implement the feature by adding every component step by step in the .js file; refreshed the My Registration page and found the registered events were gone; she switched back to the Code and Chat; copied and pasted the .js file to the AI as prompt and uploaded the PDF design mockup; double-checked with AI by asking what it made and what feature it implemented; refreshed the page and checked the update feature; tested the feature end-to-end by using the feature and checked back against the PDF wireframe design to make sure it works like the feature;

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Task 1 (Code and Chat: text prompts) - skimmed the INSTRUCTIONS.md and opened the Figma wireframe; copied and pasted the text contents from the first UI screen wireframe and the srs.md to the AI and asked the AI to find the problem, copied and pasted the second UI screen's text content to the AI for more contexts to find the mismatch; copied and pasted the text contents for the last two screen from the Figma to the AI and asked it to find more mismatches; copied and pasted the REPORT1.md file

to ask the AI to fill out the problems found; after the AI generated the table, he copied and pasted each item in the row one by one into the table for each problem; while filled out the rows, he messed up the format, he decided to copy and paste the REPORT1.md file to the AI again and asked it filled the table following the format and give him the code;

Task 2 (Code and Chat: text prompts) - skimmed the INSTRCTIONS.md file and followed the instructions to start the server and run the app; copied and pasted the 10 heuristics table to the AI and described the example (in the REPORT2.md file) problem and two Register Now buttons in the event detail page to the AI regarding the heuristic violation; while recording the problems, he interacted with the app and used the features (e.g., browsed the event detail page, About and FAQ); wrote down the third usability problem he thinks it might be (there is no Back button when clicked into the event detail page) to the AI; went back to those three reported problems and marked them with which heuristics they are violating while checking the heuristics back and forth; reported the last problem (with violating H8) of can't register for more than 1 person for one event; copied and pasted the REPORT2.md table format to the AI and asked it to fill out the problems found; copied and pasted the table into the report;

Task 3 (Code and Chat: text prompts) - skimmed the INSTRUCTIONS.md file and followed the instructions to start the server, run the application, and registered events; searched the browser to ask how he can pass the content of the PDF (images inside) without allowing the images uploaded; prompted the AI the Background section information from the INSTRUCTIONS.md and informed that he can't upload the images and will provide the code later; copied and pasted the MyRegistrationView.js and components.css to the AI and asked it to apply the change on the feature from the (Background info he passed before); copied and pasted the updated code to the files and registered an event to see the update; he found it works without further checking or using the feature;
